

REVISION HISTORY

Revision No./Version No.	Date	Prepared by	Approved by	Details of Change
1.0	10-Jul-2024	Siya Chavan (Senior TestLead)	Anant Sham Yende (CTO)	Test Management SOP

OBJECTIVE

The purpose of this SOP is to establish a standardized and structured approach for developing test cases to validate software functionality, test case execution, and defect management to ensure delivery of high-quality software products.

SCOPE

This SOP applies to all testers, and technical teams involved in the development and testing of software product within the organization. It involves in identification of test requirements, test scenarios, designing detailed test cases, execution of test cases, validating software functionality and performance, identification, prioritization, resolution and verification of defects to ensure the quality of the software product, documenting defects and obtaining formal approval to conclude the testing phases.

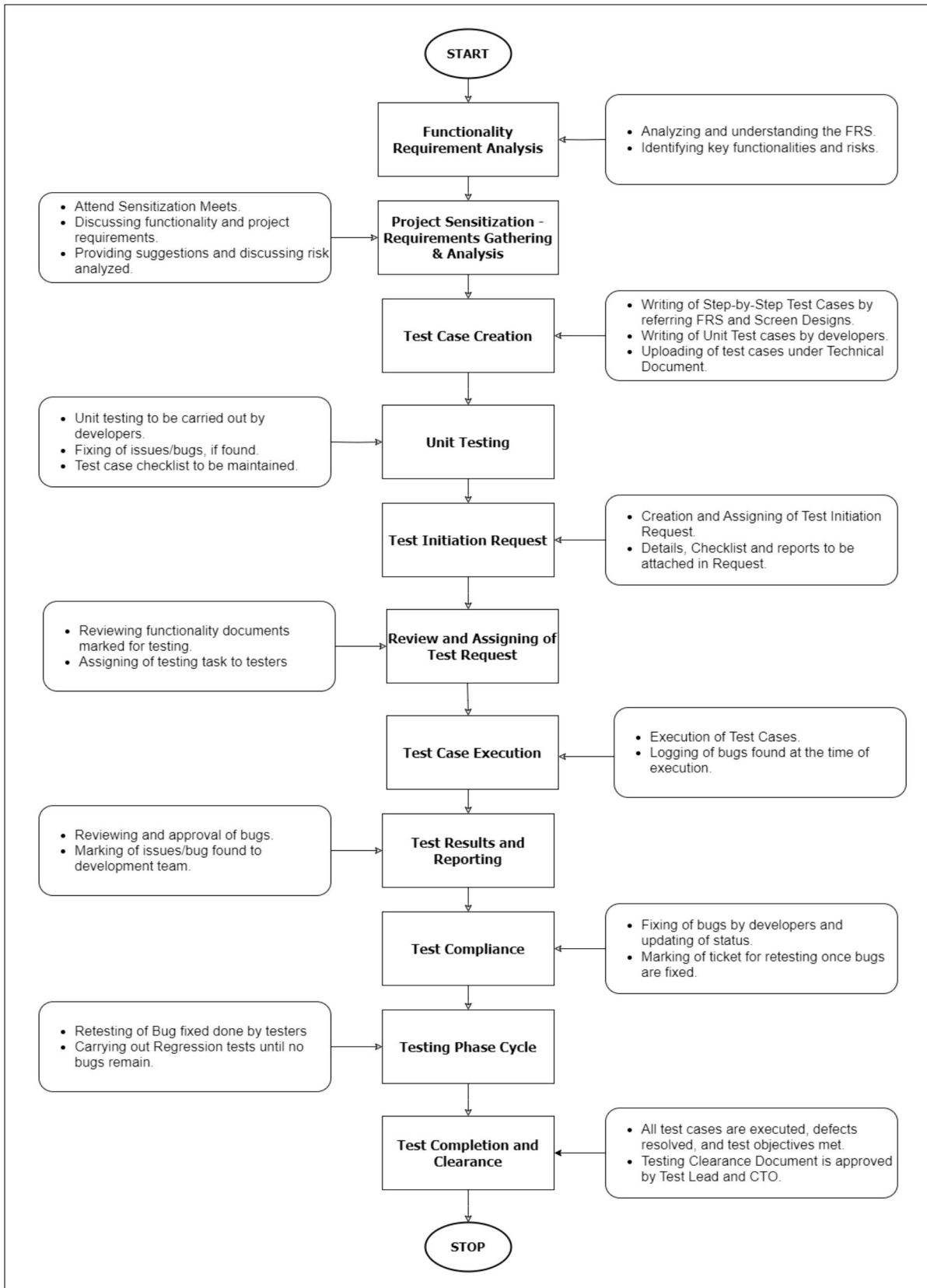
PARTICIPANTS

1. **Senior Test Lead:** Assign Testing tasks, Coordinates test execution activities and monitors progress, verifying defects and ensures their alignment with requirements. Reviews test results and prepares test closure report.
2. **Testers:** Create, execute, and maintain test cases and test results. Log and Prioritize defects through out the test life cycle.
3. **Project Managers:** Oversee the development and testing process to ensure alignment with project goals. Provide inputs related to the requirements and functionality to be tested.
4. **Developers:** Provide input on test scenarios at the time of preparation of test cases. Investigate and resolve reported defects in timely manner.
5. **Business Analysts:** Provide requirements and clarify functionality to be tested.

PREREQUISITE

1. **Requirement Documentation:** Up-to-date requirements documents (e.g., Software Requirements Specification (SRS)/ Functional Requirement Specification(FRS), Screen Designs, Test Initial Request)
2. **Tools and Software:** Defect Tracking/Management in GELJIRA configured and integrated.
3. **Team and Training:** Defined Roles and Responsibilities; Training Sessions on the Test Management and Defect Tracking tool.; Training on defined Validation Matrix and Test scenarios.
4. **Standards and Guidelines:** Established Standards and templates for writing test cases.; Defined review and approval process for test cases.

TESTING WORKFLOW



1. FUNCTIONALITY REQUIREMENT ANALYSIS

- Developers and Testers to review requirements documentation (FRS/SRS/Screen Designs) to understand the scope, functionality, features, and expected behavior of the software.
- To identify key functionalities, features, risks and challenges involved in development or testing process.

2. PROJECT SENSITIZATION - REQUIREMENT GATHERING & ANALYSIS

- Technical Team along with Testers should attend Project Sensitization Meet.
- Developers and Testers should discuss key features, functionalities and risks identified that need to be developed/tested and also provide suggestions and asks queries if any, with respect to the form/flow/design/functionality of the module.

3. TEST CASE CREATION

3.a. **Define Test Cases:** Write test cases using the test case template (Annexure I), which includes:

- Requirement ID: Unique identifier of the requirement being tested.
- Test Case ID: Unique identifier for each test case.
- Test Case: Descriptive title that summarizes the test case.
- Test Steps: Detailed, step-by-step instructions for executing the test.
- Expected Results: The expected outcome of the test case.

3.b. **Review and Approval:**

- Tester to review the first level test cases for accuracy, completeness, clarity and relevance.
- Tester to validate each test case against the corresponding requirement.
- Incorporate feedback and get approval from the Test Lead or Senior Test Lead.

3.c. **Maintain Test Cases in GELJIRA:**

- Upload the test case towards the project in GELJIRA towards respective Development request or change request testing ticket.
- Regularly update test cases to reflect changes in requirements or system updates.
- Archive obsolete test cases and maintain a version history.

4. UNIT TESTING

- Developers write unit test cases and tests to check individual units of code (e.g., methods, functions).
- Developers to run the tests on the code and fix any issues that arise.
- Developers should go through the test cases written by testers and ensure they align with the functional requirements and are comprehensive.
- Developers to use the test cases as a checklist to make sure their code meets the requirements outlined in the test cases.

5. TEST INITIATION REQUEST

- After developers complete the coding and initial unit testing of the software, the Technical Lead should create Test Initiation Request in GEL JIRA.
- Test Initiation Request should include relevant details required for testers.
- Test Initiation Request should be attached with Unit Test Report and Test Case Checklist report which is done by the developers. Test Cases Checklist Report is created to confirm that developers have reviewed and addressed each of the test cases in their code
- Once all the details and reports are uploaded and build is ready, the Technical Lead should assign the ticket to the Test Lead, signalling that testing can begin.

6. REVIEW AND ASSIGNING OF TEST REQUEST

- Test Lead should review the Details uploaded in the Test Initiation Request.
- Test Lead should verify the Unit Test Report and Test Case Checklist Report uploaded by developers, and if any issues or missing details, then should communicate back to the development team if necessary.
- Test Lead assigns the Testing Ticket in GELJIRA to specific testers based on their roles, expertise, or the module they are responsible for testing.

7. TEST CASE EXECUTION

6.a. Preparation:

- Ensure the test environment is set up and meets preconditions.
- Verify the availability of the latest build or version of the software.

6.b. Execution:

- Follow test steps as defined in the test case.
- Record actual results and update the status (Pass/Fail/Blocked).

6.c. Defect Reporting:

- Log defects as 'Testing Bug Issue' in GELJIRA as Sub Tasks towards the corresponding Testing Ticket.
- Providing detailed information about the defect, including:
 - Issue Summary: Describe the observed behavior or issue which gives gist of issue.
 - Issue Sub Type: Select type of bug (Functional/Validation/Report/UI).
 - Steps to be Performed: Provide Clear, Step by Step instruction to reproduce the defect.
 - Expected Result: Specify the expected outcome or behavior.
 - Actual Result: Describe the actual outcome or behavior observed.
 - Attachments: Include relevant screenshots, logs, or other supporting documentation.
 - Priority: Assign appropriate priority/severity level to the bug based on the predefined criteria.
- Provide Suggestions/Enhancements in GELJIRA w.r.t. the software while execution.
- Add queries in GELJIRA w.r.t. the software if incase software behaves in way which is not documented.

8. TEST RESULT REVIEW AND REPORTING

7.a. Review Execution Results:

- Ensure accuracy of the test execution results.
- Validate and retest failed test cases if necessary.

7.b. Reporting:

- Generate test execution reports summarizing testing progress and results.
- Mark the Testing Ticket to Development team after Testing Phase is completed and Test Execution results are verified.
- Testing Clearance Certificate is Issued after testing is completed and no bugs found.

9. TEST COMPLIANCE

- The development team works on the issues identified during testing and once bugs are resolved, developer should update the status of each bug accordingly. After bug is fixed, developer needs to ensure it is not replicated again on test environment.
- Once all the bugs are fixed by the development team and Bug Tickets are closed, the Technical Lead assigns the Testing ticket to Test Lead marking the status as 'Phase-1 Issue Resolved'.

10. TESTING PHASE CYCLE

- Once Compliance of Testing Ticket is received, Test Lead assigns the ticket to the Tester and Tester should start with Test Execution.
- Testers should retest all the bugs found in previous phase of testing and should also run regression tests to ensure that new fixes or changes have not introduced new bugs in other areas of the module.
- After Test execution is completed and when issues are found, the same process of Test Result and Reporting, Test Compliance continues. This cycle is repeated, increasing the testing phase number (Phase-2, Phase-3, etc.) until no more bugs are found.

11. TEST COMPLETION AND CLEARENCE

- After completing all testing phases and finding no remaining bugs, the testing team prepares the Testing Clearance Document.
- The Test Lead and CTO review and approve the Testing Clearance Document.
- The clearance document is uploaded to the Testing Ticket.
- The Test Lead assigns the ticket back to the Technical Lead as Phase-n Testing Cleared, marking the end of the testing process, signaling that the software/module is ready for deployment or release

STANDARD OPERATING PROCEDURE

Test Management

DOCUMENTATION

1. **Test Case Document:** Detailed document containing all test cases, structured as per the Annexure I.
2. **Test Execution Reports:** Reports summarizing the results of test case execution, including pass/fail status and defect details.
3. **Defect Reports:** Documentation of all logged defects in GELJIRA.
4. **Testing Clearance document:** Documents which includes Project and Testing task details, Types of testing performed, Testing Phase Summary, Exception issues counts and details, Remarks and Approval Signatures of Test Lead and CTO.

ANNEXURE I

Requirement ID	Test Case ID	Test Case	Test Steps	Expected Results